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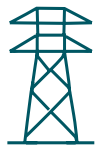
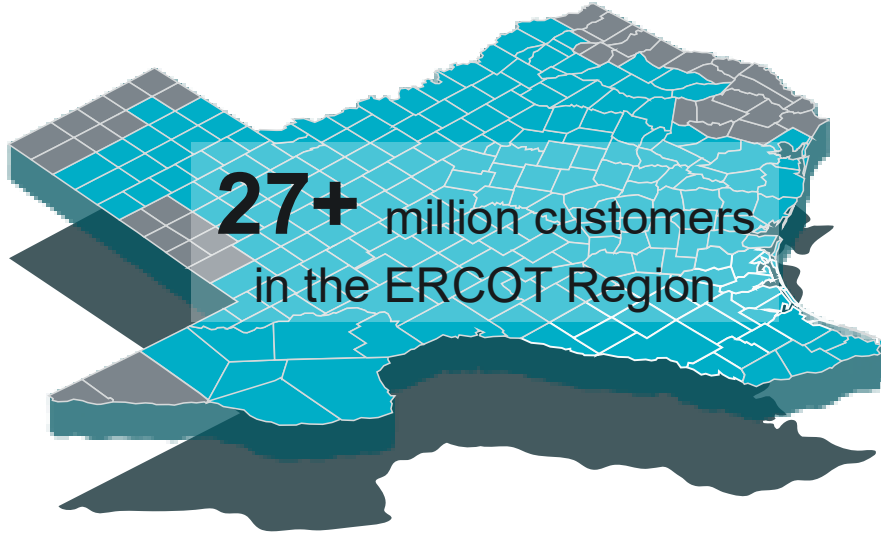
Predicting the Unpredictable: Forecasting Flexible Large Loads in ERCOT

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ERCOT Facts



~\$14 billion Transmission projects endorsed in 2025



1,460+ generating units, including PUNs



1,928+ active Market Participants that generate, move, buy, sell, or use wholesale electricity

Summer 2026 Installed Capacity

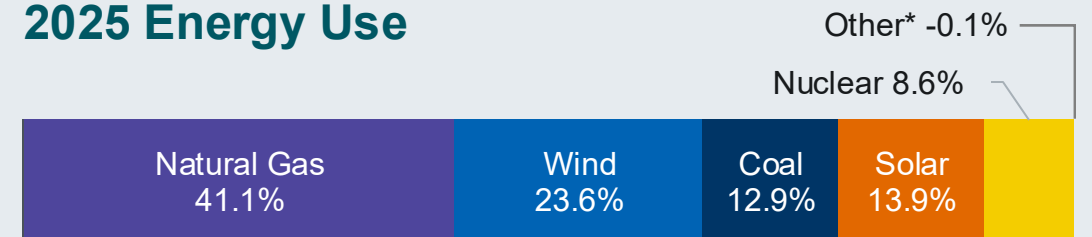
Reflects operational installed capacity for Summer 2026 based on `December 2025 Capacity, Demand and Reserves report.



The sum of the percentages may not equal 100% due to rounding.

*Other includes biomass and Direct Current Tie capacity.

2025 Energy Use



*Other includes solar, hydro, petroleum coke (pet coke), biomass, landfill gas, distillate fuel oil, net DC-tie and Block Load Transfer imports/exports and an adjustment for wholesale storage load.

ERCOT Records

**Peak
Demand
Record**

85,508 MW
(August 10, 2023)

**Weekend
Peak Demand
Record**

85,116 MW
(August 20, 2023)

**Expected Capacity
for Summer 2026**

104,850+ MW
(December 2025 *Capacity,
Demand and Reserves report*)

1 MW of electricity is enough to serve about **250** residential customers during ERCOT peak hours.

Wind, Solar & Energy Storage by the numbers



40,739 MW

of installed wind capacity as
of May 2026, the most of any
state in the nation

29,011 MW

Generation Record
(May 18, 2026)

69.15%

Penetration Record
(April 10, 2022)



39,365 MW

of utility-scale installed
solar capacity as of
May 2026

34,498 MW

Generation Record
(May 13, 2026)

72.74%

Penetration Record
(May 2, 2026)



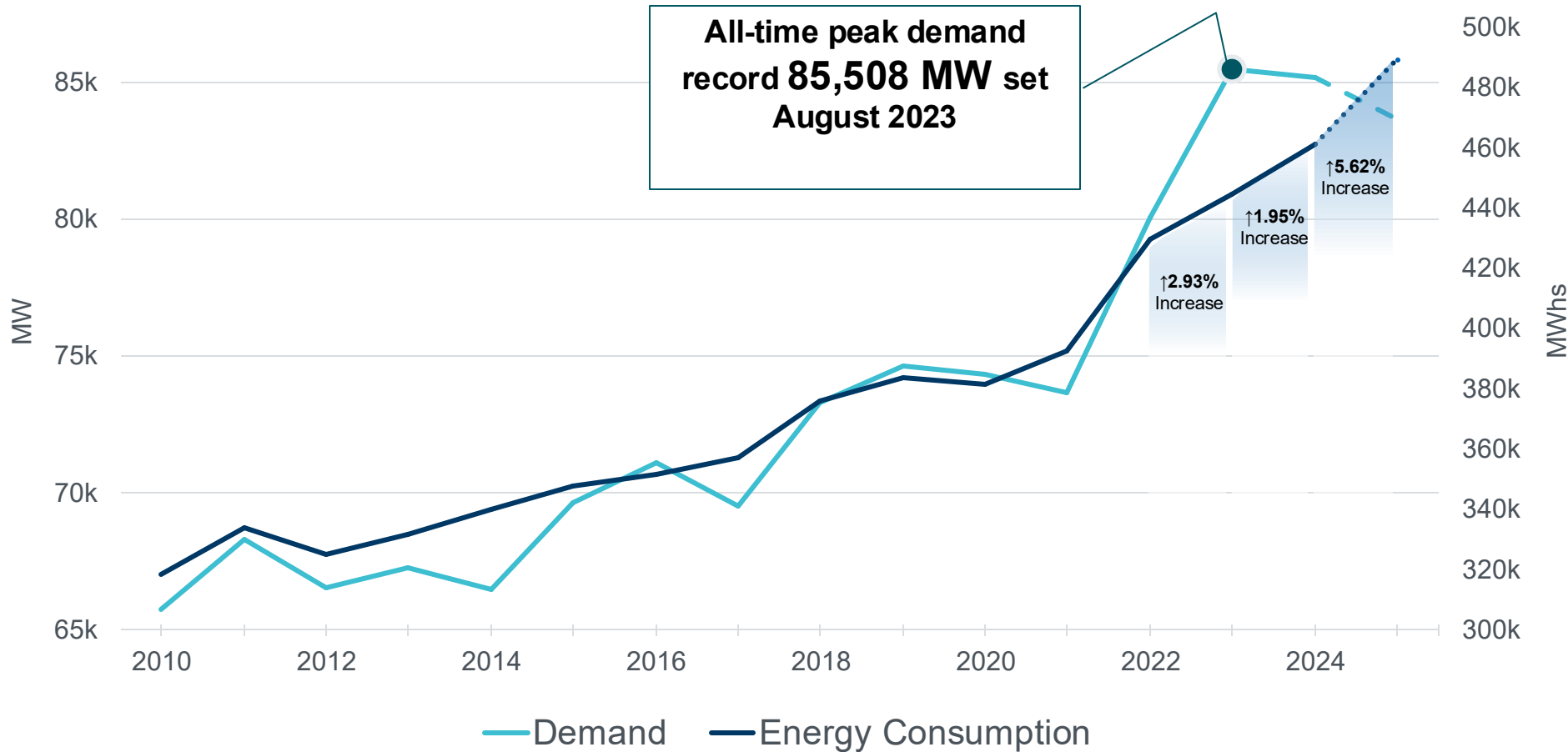
20,418 MW

of installed energy storage capacity as of
May 2026

10,515 MW

Energy Storage Generation
Record (March 13, 2026)

ERCOT: Growth in Peak Demand and Energy Consumption



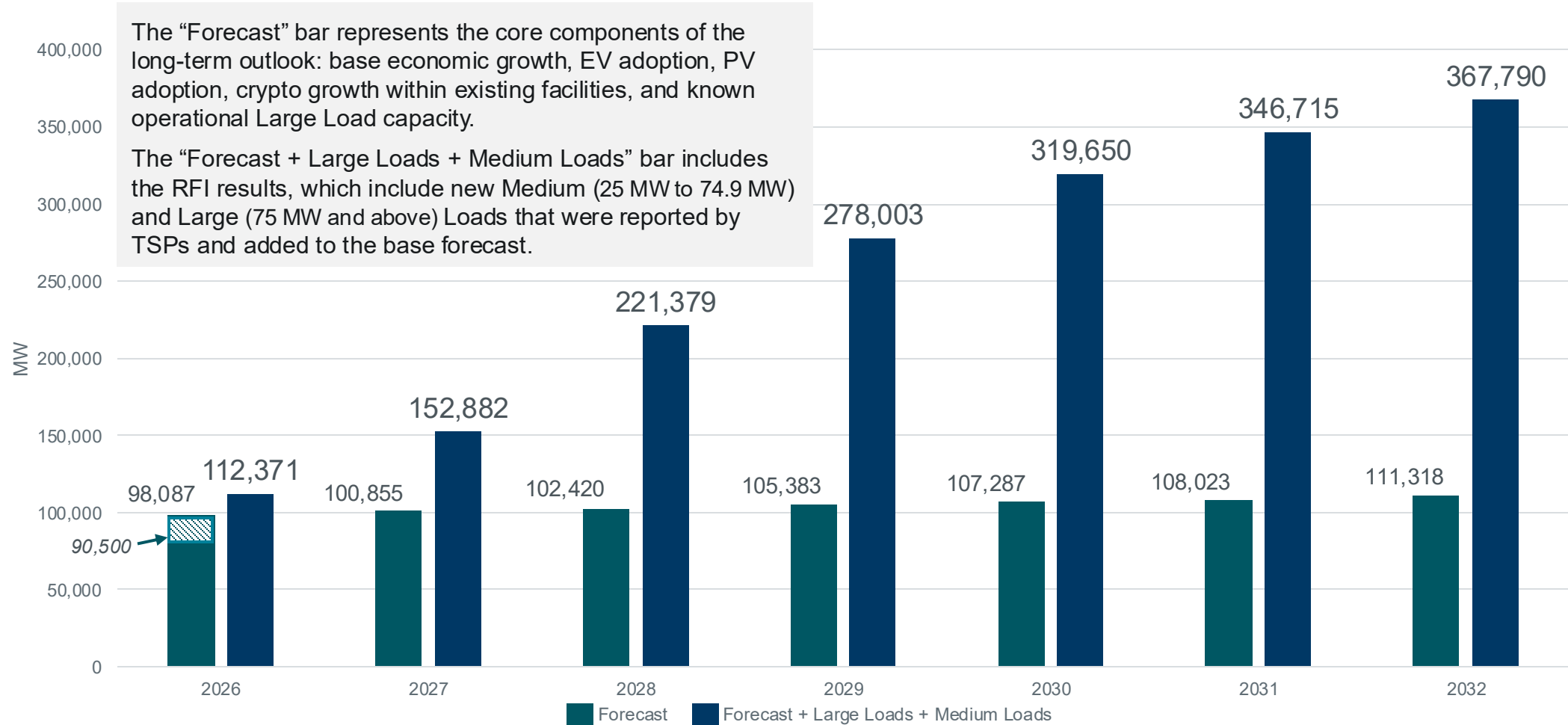
Average Energy Growth

Full Period	Average Energy Growth
2010-2025	2.93%
2010-2020	1.95%
2021-2025	5.62%

This is settled load data and does not include the ESR's.

Key Takeaway: While ERCOT hasn't set a new all-time peak demand record since August 2023 (85,508 MW), energy consumption continues to increase year over year.

Preliminary Long-Term Load Forecast Peak Demand plus Transmission Service Providers (TSPs) Request for Information (RFI) Submissions

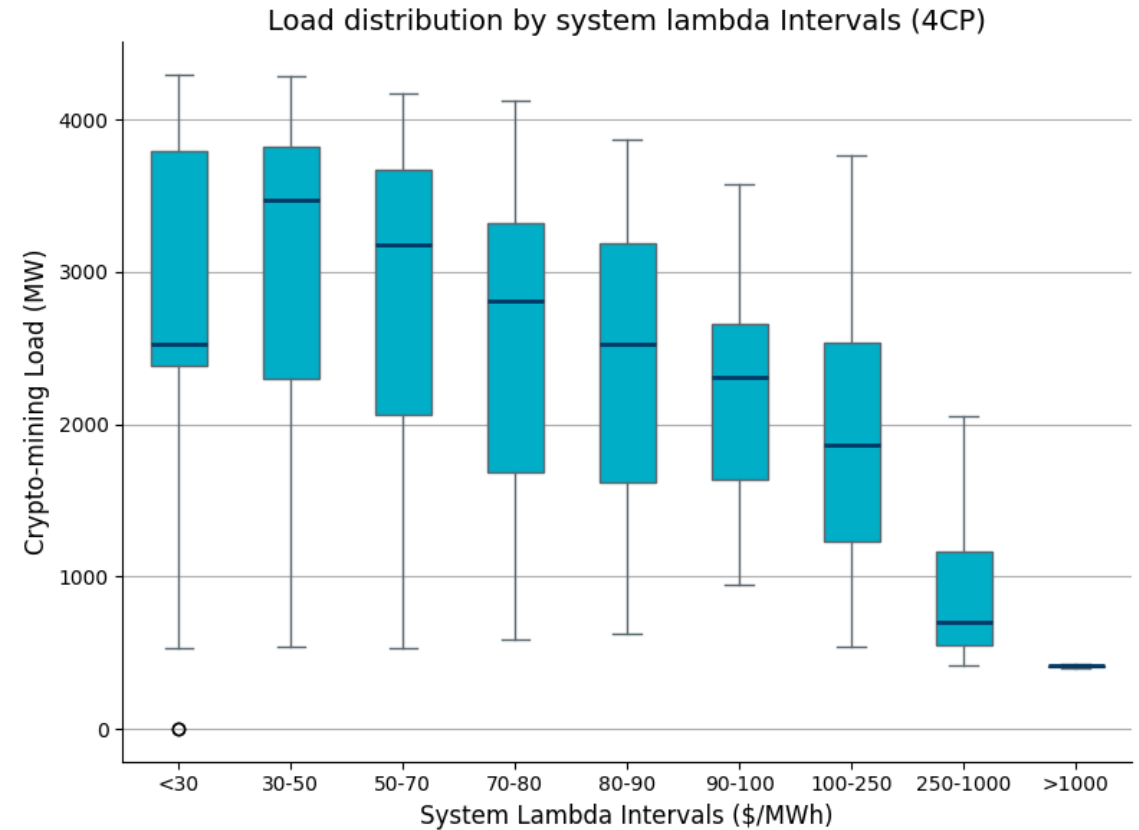
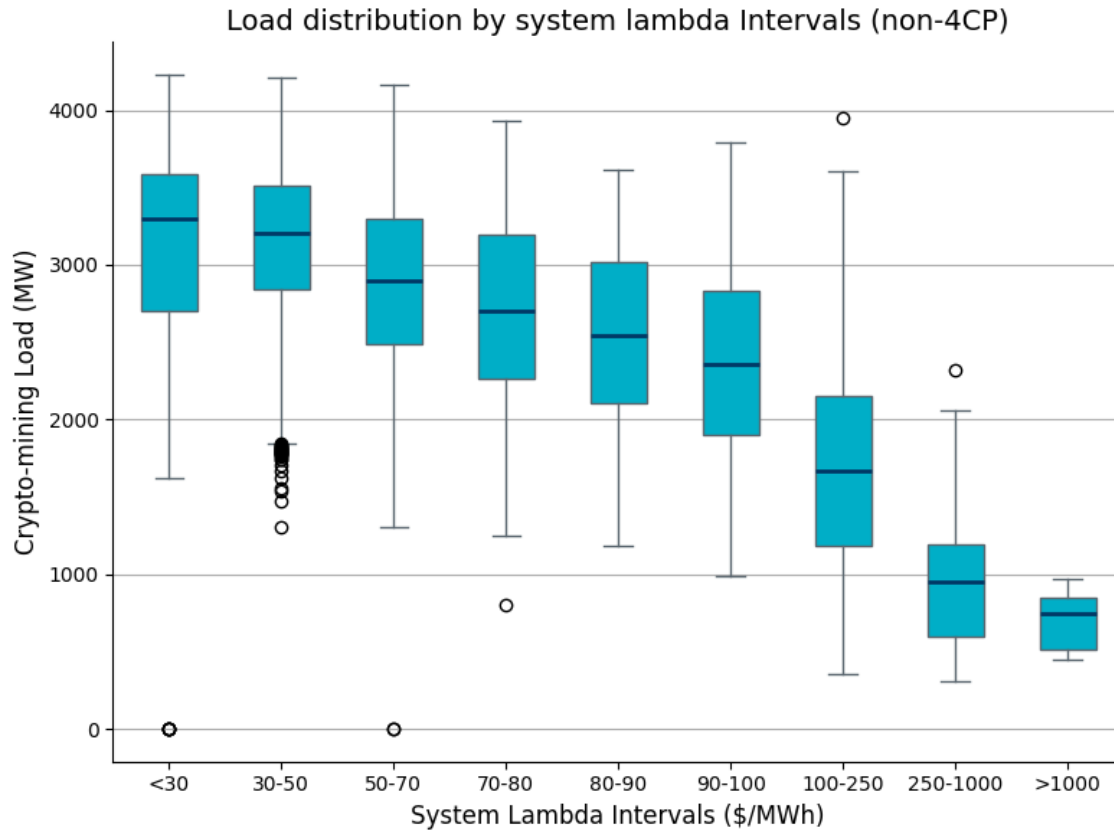


Key Takeaway: This forecast should be considered preliminary, and adjustments are expected as the information is further studied, reviewed, and finalized to support its use in planning.

TSP Large Load RFI Submissions by Load Type

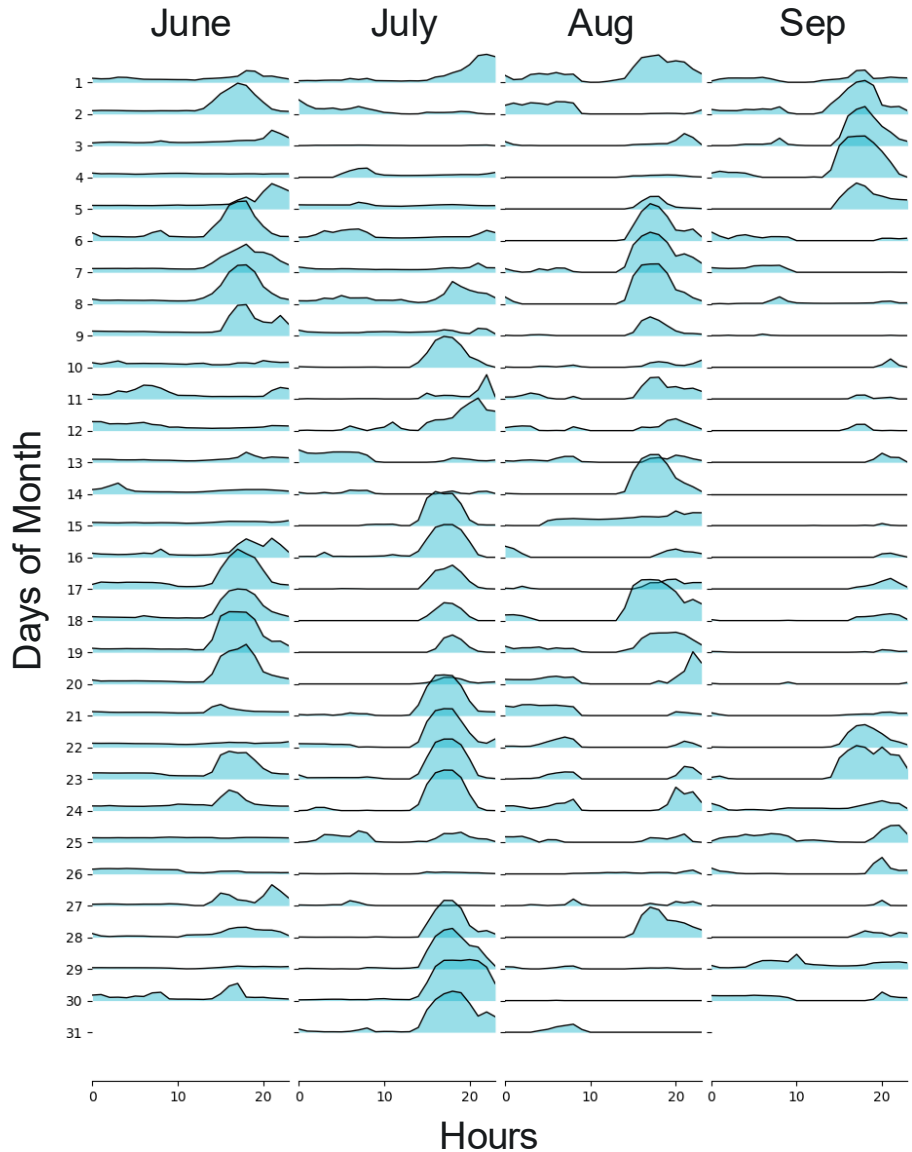
Load Type	2026	2027	2028	2029	2030	2031	2032
Cryptocurrency Mining	2,475	3,518	5,611	7,126	7,676	8,576	9,076
Data Center (non-crypto)	7,401	39,719	99,350	149,258	186,958	211,624	228,420
Industrial (incl. Hydrogen/E-Fuels)	650	1,080	1,805	2,259	2,721	2,919	2,996
Oil and Gas (Chemical Refining, Production, and Processing)	917	1,214	1,650	2,017	2,187	2,347	2,507
Total	11,443	45,531	108,416	160,660	199,542	225,466	242,999

Crypto-currency Mining Load Behavior



Key Takeaway: Crypto-currency mining load behavior is impacted by two major factors – price response and four coincidence peak avoidance. During summer months, large demand reduction could occur even if system price is low.

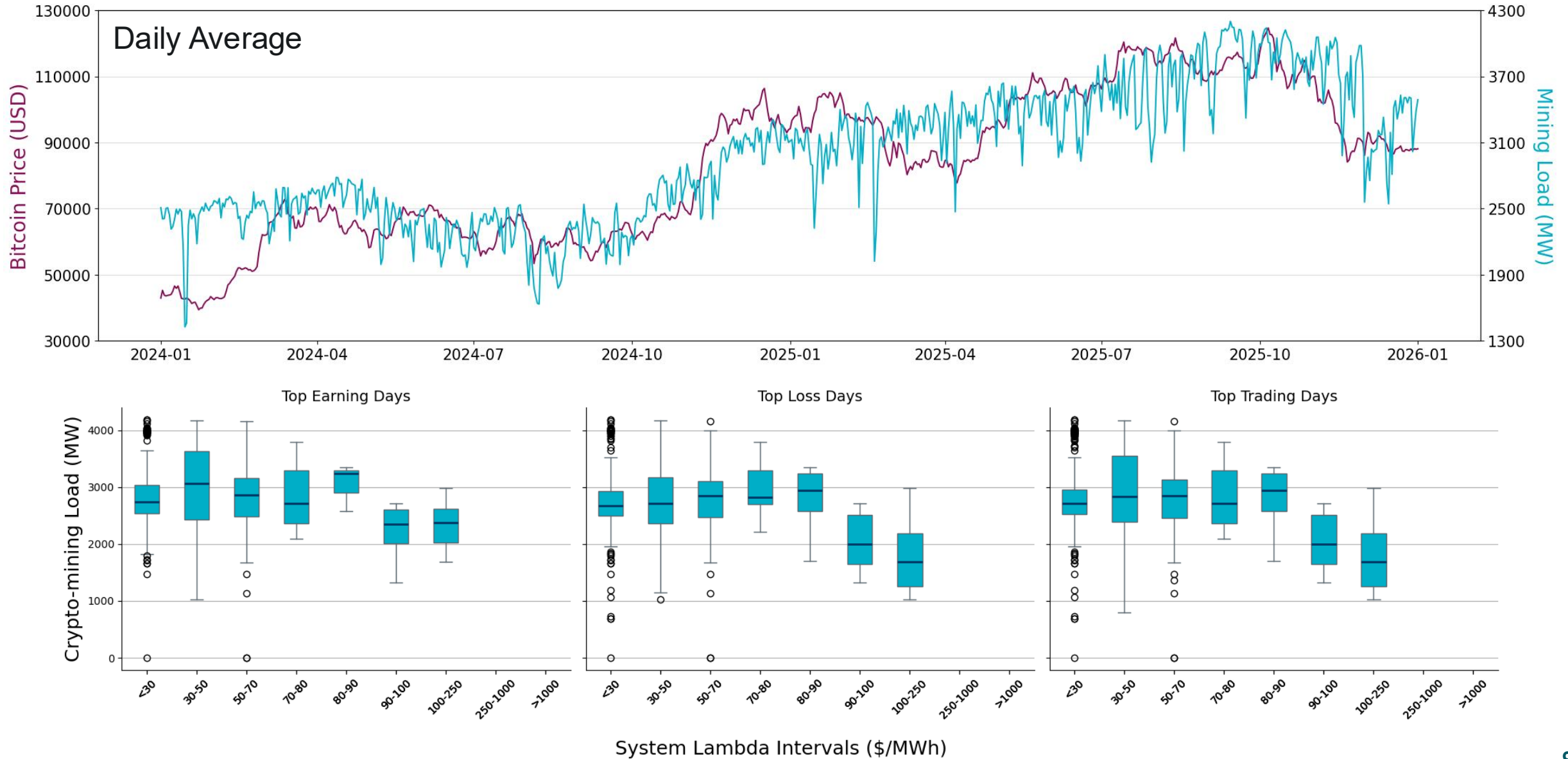
Crypto-currency Mining Load Behavior (4CP)



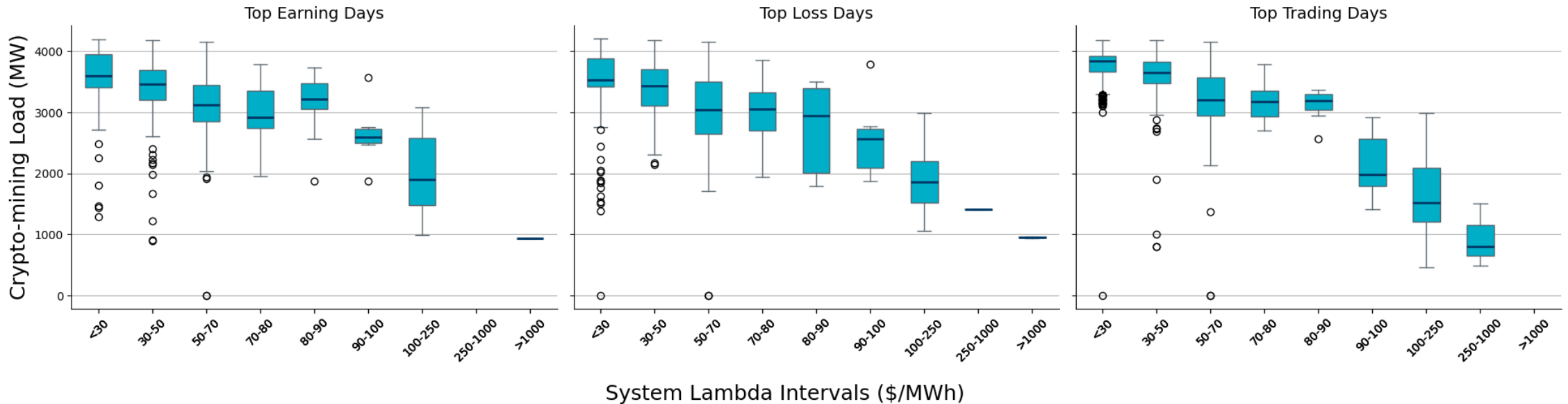
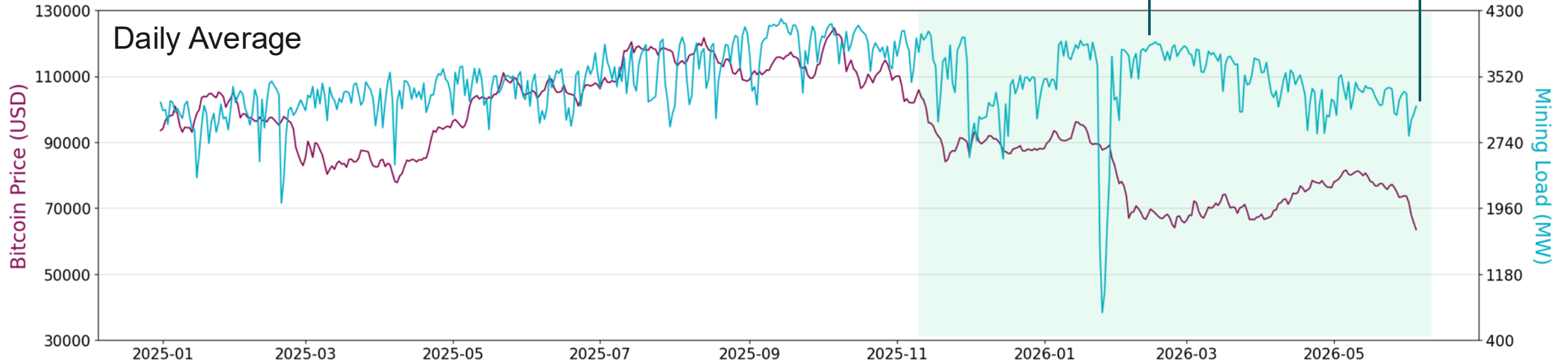
- Load reduction amount per day across summer months, higher = more curtailment
 - 2025 June, July, August, and September
- Load reduction patterns remain consistent on actual 4CP avoidance days
- Large renewable penetration pushes price peaks towards net load peaks but 4CP is determined on load peaks
- Battery charging introduced variability in 2024 and 2025 (pre-RTC): 4CP uses settled load (e.g., load - battery charging MW), while real-time dispatch operates on telemetered native load

Key Takeaway: Different models are needed to capture both price-responsive and 4CP behaviors

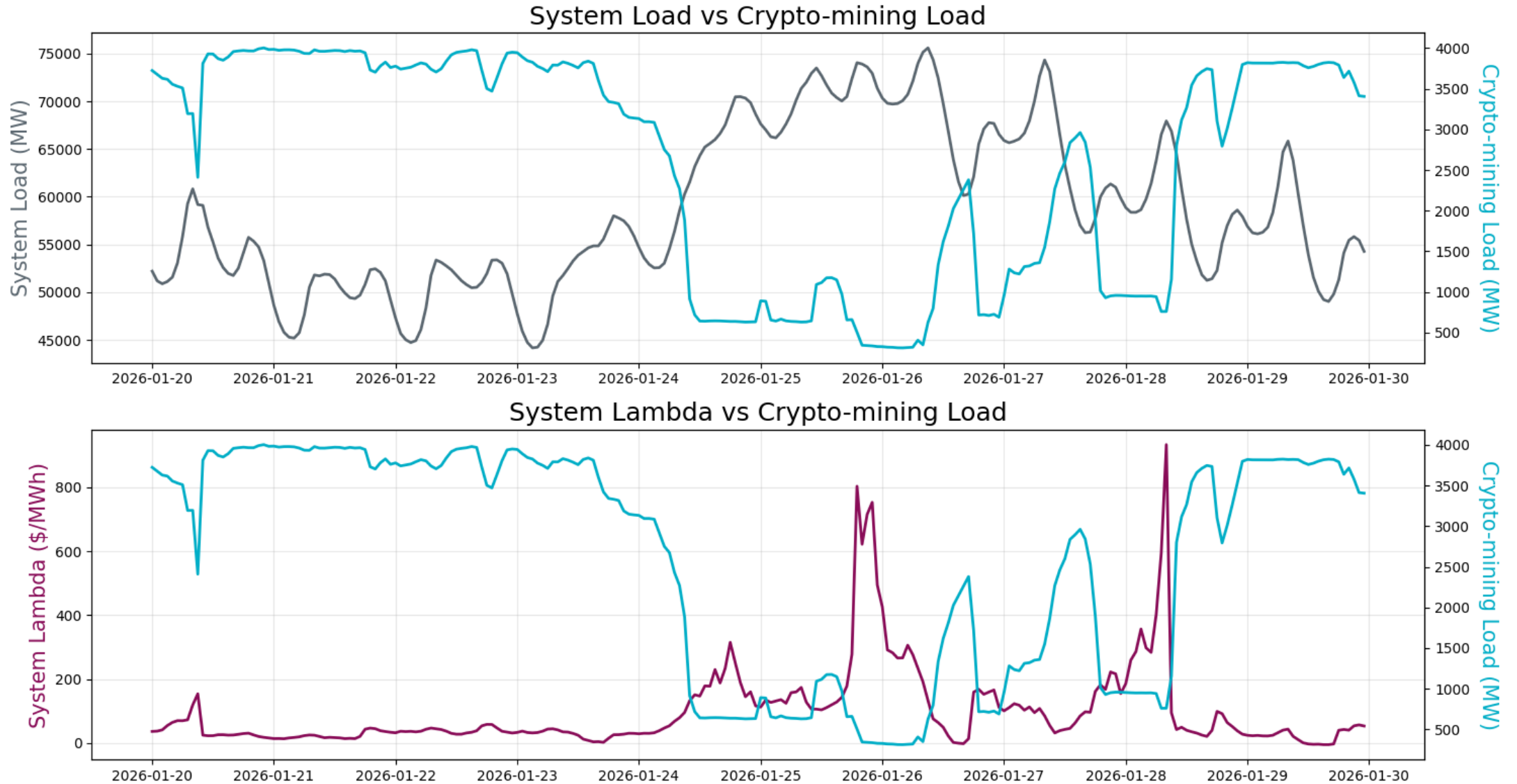
Bitcoin Price and Load Consumption Correlation



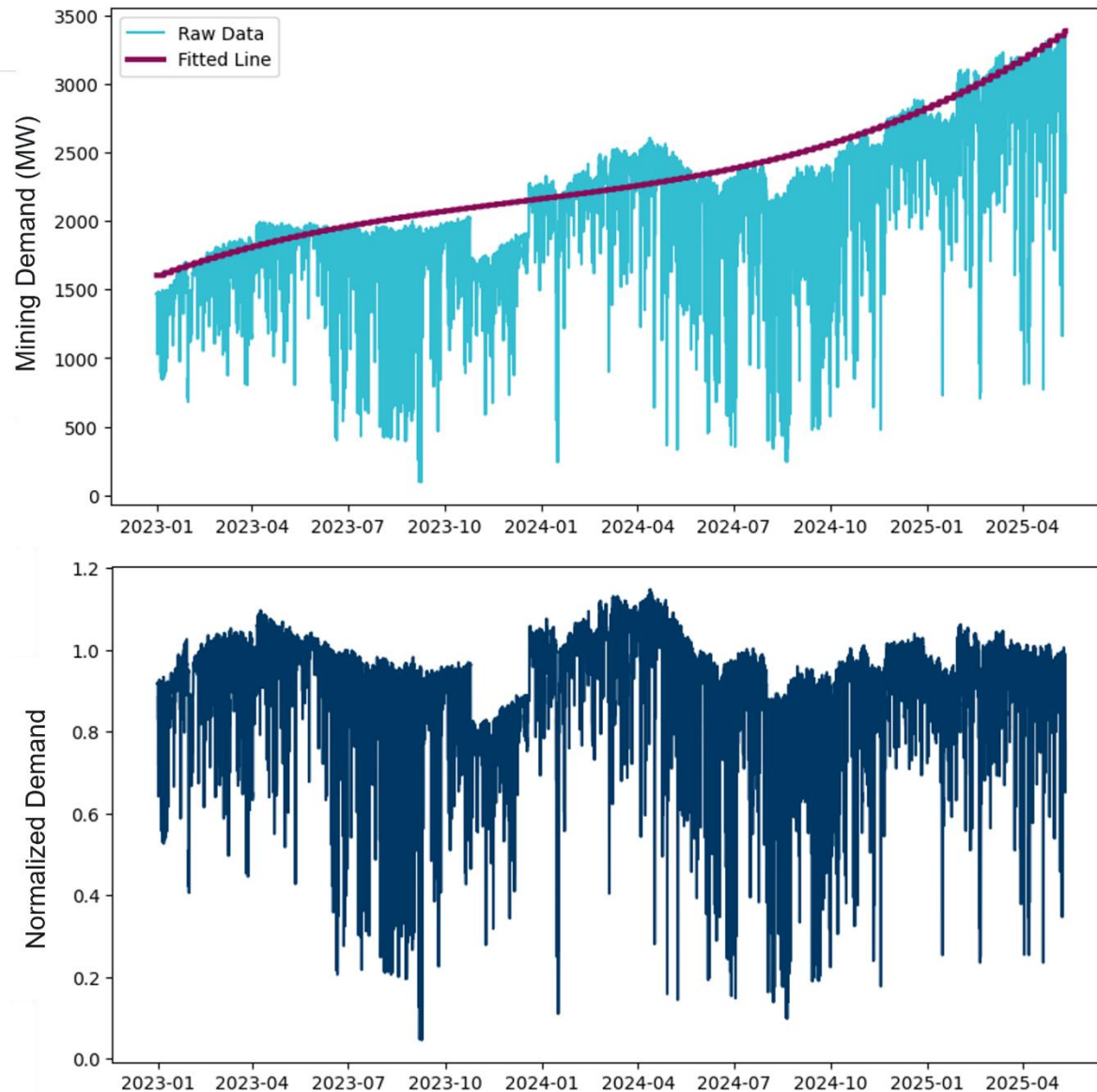
Bitcoin Price and Load Consumption Correlation



Crypto-currency Mining Load Observed Winter Storm Fern Behavior

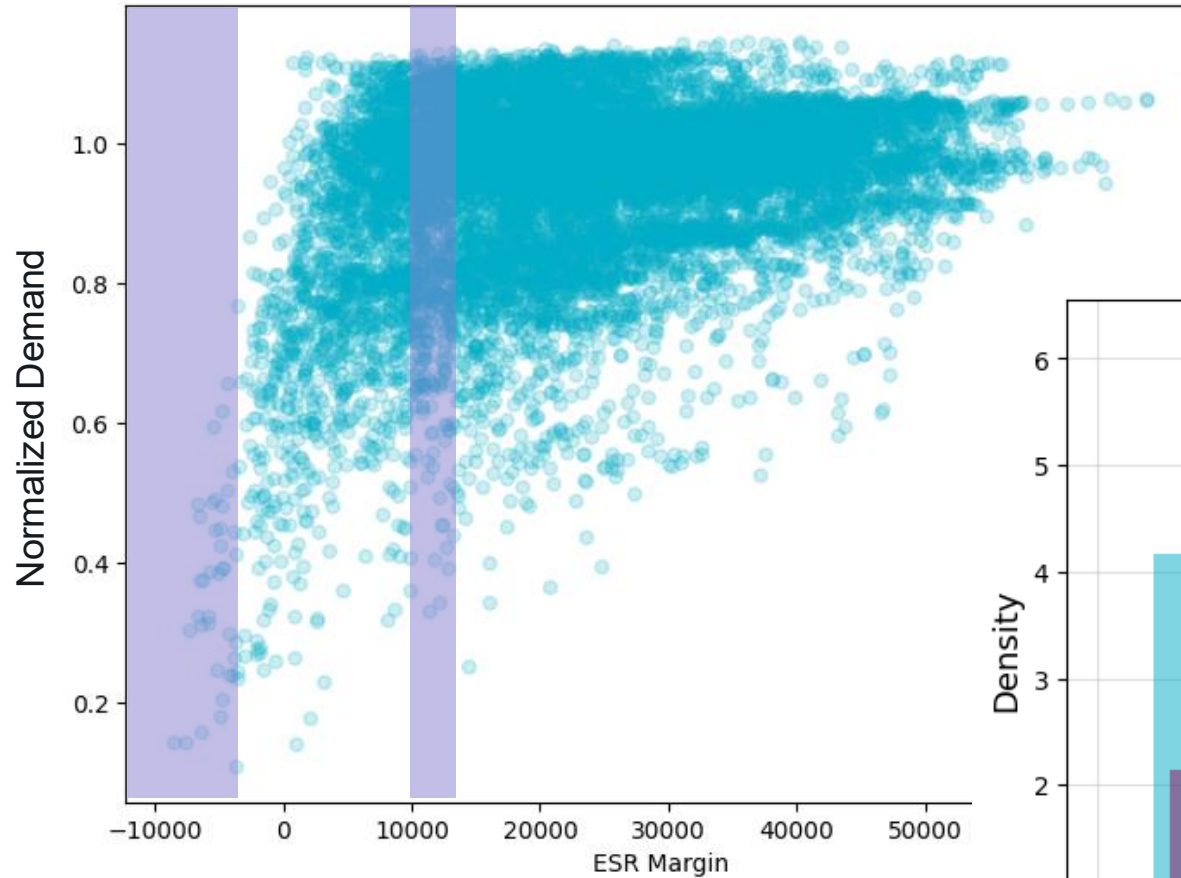


Large Load Forecasting – Statistical Model

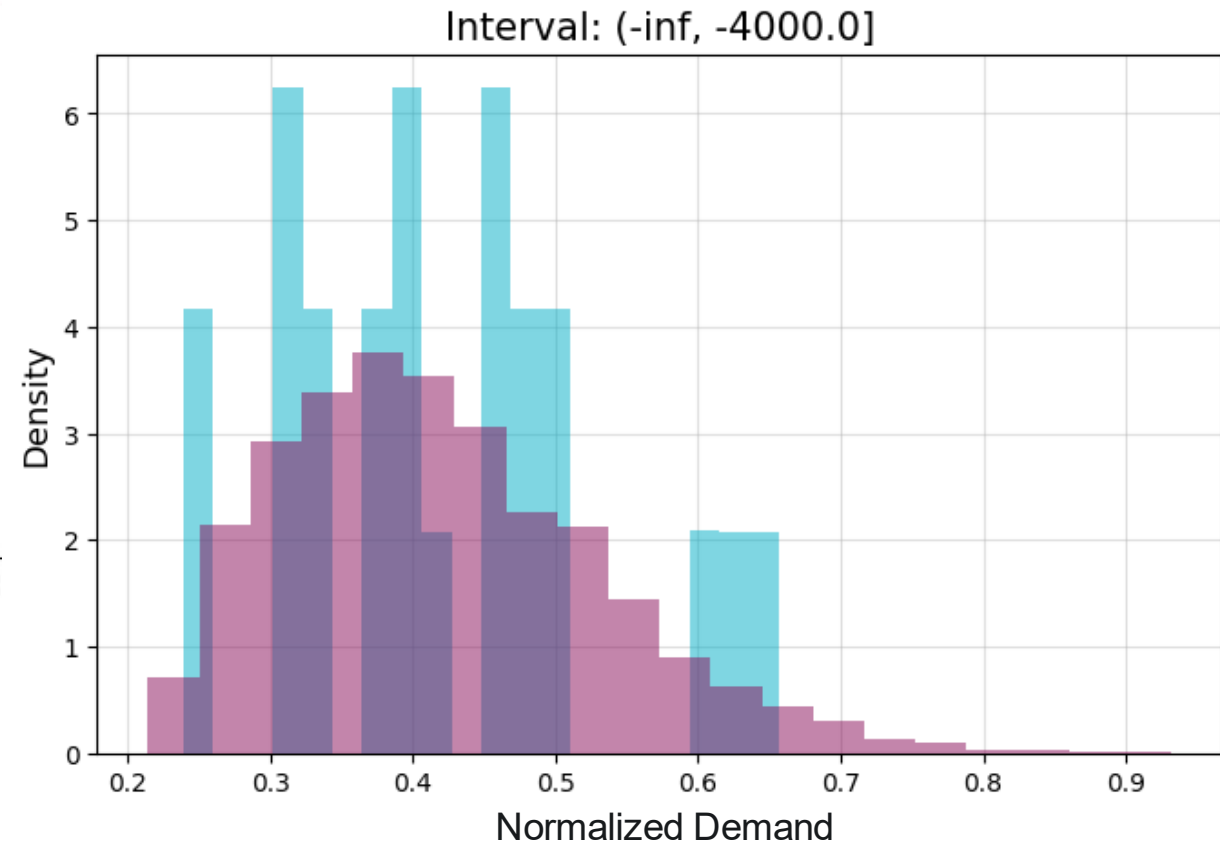


- Growing demand leads to more volatile reduction
- Need to be careful how to detrend overall demand
- Energy storage resource (ESR) margin is calculated and used as price signal proxy
 - $\text{ESR margin} = \text{Thermal capacity} + \text{solar} + \text{wind} - \text{forced outage} - \text{planned outage} - \text{load}$
 - Estimation of how much ESR capacity is needed

Mining Load Forecasting – Statistical Model

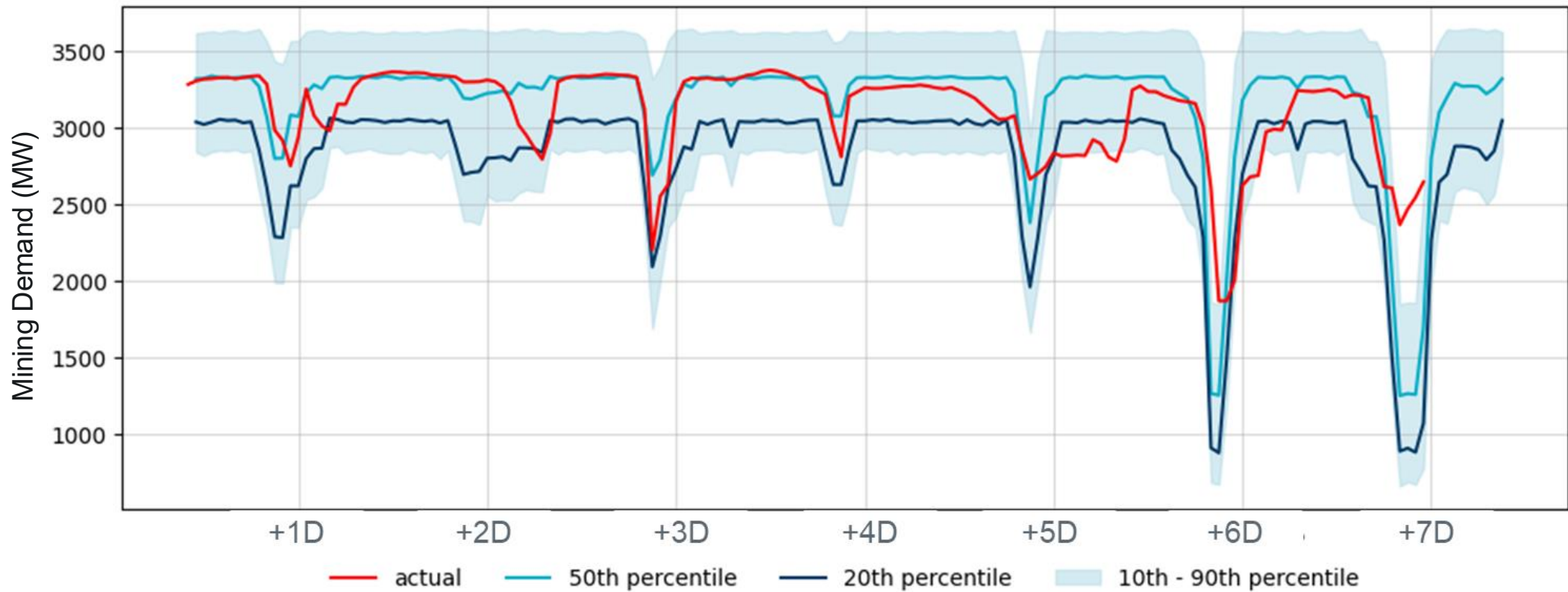


- January 2023 – May 2025
- 4CP months vs non-4CP months

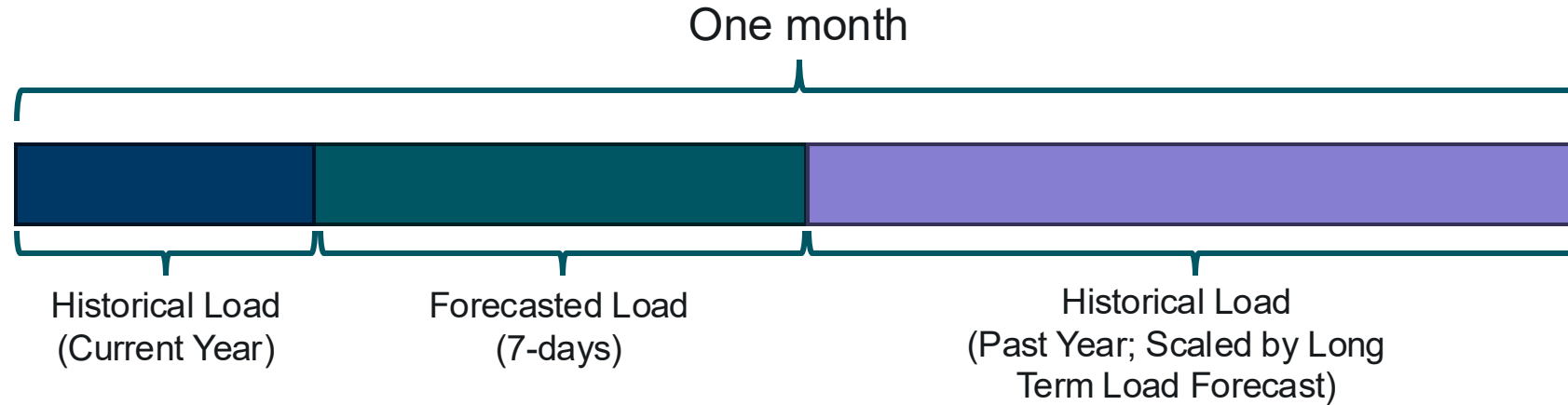


Mining Load Forecasting – Statistical Model

- Apply multiplier based on recent load magnitude
- ESR margin calculated using probabilistic forecasts for wind, solar, load, and forced outages
- Example 168-hour snapshot:

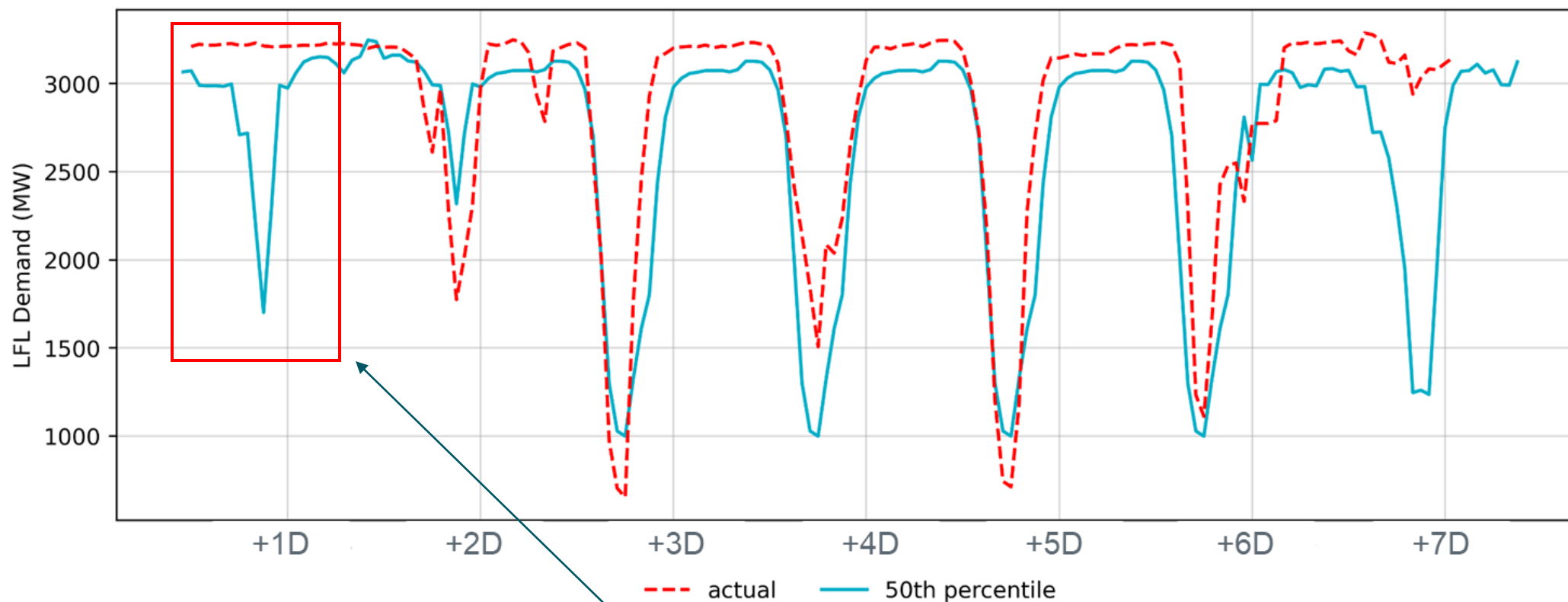


4CP Prediction



- For 4CP prediction, we combine actual load from current month/year, the latest load forecast, and the historical demand from the previous year
 - Historical load is scaled using long term load forecast to account for growth in demand
 - The highest 11 load days are calculated from this window
- After the initial forecast has been generated, if any of the days in the forecast horizon is one of the top 11 days, a new demand curve is generated from the responsive 4CP distribution for that day

Mining Load Forecast Results – 4CP



Room for further improvements

Next Steps / Challenges

- Improvement of mining load forecast model
 - Training separate models for 4CP month and non-4CP months
 - Classification / regression hybrid model
- Challenges
 - Continue shift in behavior
 - More mature 4CP estimation methodology
 - Re-tooling from Bitcoin mining to data centers
 - Lack of unit-level details for better forecast from bottom-up
 - Price forecasting
 - Capacity scarcity vs congestion

Questions/Comments?

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